MTH 336(A) Dr. James Ashe
Worksheet 10.2, 11.1-11.3 Monday 16$^{\text{th}}$ September, 2013 **Name:**

Determine the radius of convergence of the following power series.

1. $\sum \frac{x^k}{5^k}$

Use the geometric series $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$, *for* $|x| < 1$ *to find a power series representations for the following functions, and determine the interval of convergence.*

2. $f(x) = \frac{10}{1-x^2}$.

3. $f(x) = \frac{2x^2}{1-x}$.

Eliminate the parameter in following equations to obtain a single equation in terms of x *and* y.

4. $x = 2t$, $y = 4t^2$.

Determine dy/dx *in terms of* t *and evaluate it at the given value of* t.

5. $x = 2t$, $y = 4t^2$; $t = 1$.

Give a set of parametric equations that describes the curve.

6. A circle centered at $(0, 0)$ with radius 6, generated clockwise.